

Chapter 7: Applications of Integration

Section 7.3: Volumes: The Shell Method

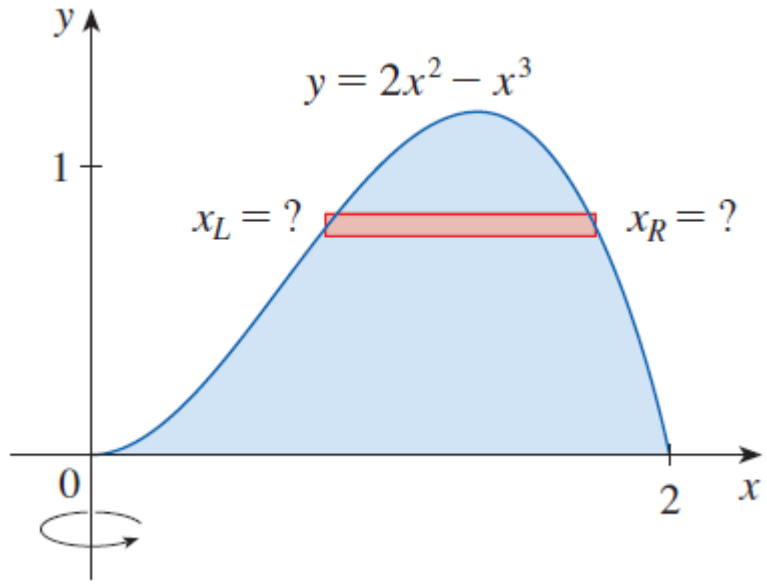
EXAMPLE

Find the volume of the solid generated by revolving the region bounded by the curves $y = 2x^2 - x^3$ and $y = 0$ about the y-axis.

The cross-sections perpendicular to the y-axis are washers.

In order to find the inner radius and outer radius, we have to solve the equation $y = 2x^2 - x^3$ for x which is not easy to do.

There is another method, called the **method of cylindrical shells**, that is easier to use in such cases.



The Method of Cylindrical Shells

A cylindrical shell is a solid between two concentric cylinders.

Figure 2 shows a cylindrical shell with inner radius r_1, outer radius r_2, and height h. Its volume V is calculated by subtracting the volume V_1 of the inner cylinder from the volume V_2 of the outer cylinder:

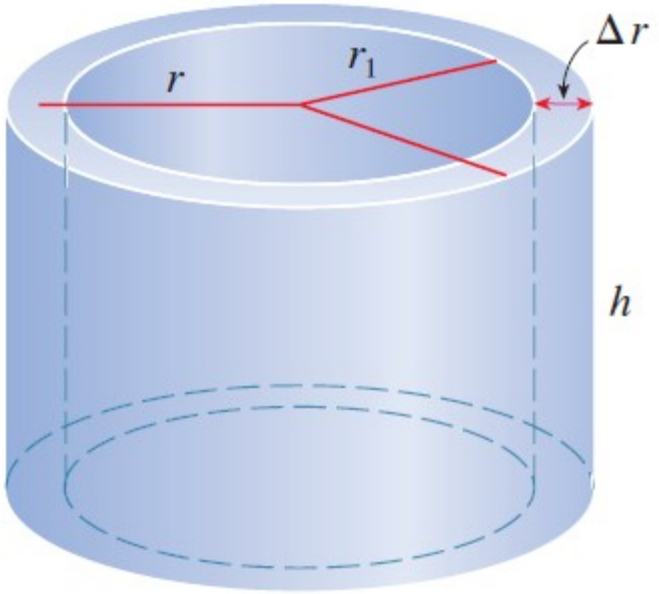


FIGURE 2

$$
\begin{aligned}
V &= V_2 - V_1 \\
&= \pi r_2^2 h - \pi r_1^2 h = \pi (r_2^2 - r_1^2) h \\
&= \pi (r_2 + r_1)(r_2 - r_1) h \\
&= 2\pi \frac{r_2 + r_1}{2} h (r_2 - r_1) \\
&= 2\,\pi\ (average\ radius)(height)(thickness)
\end{aligned}
$$

If we let $\Delta r = r_2 - r_1$ (the thickness of the shell) and $r = \frac{1}{2}(r_2 + r_1)$ (the average radius of the shell), then this formula for the volume of a cylindrical shell becomes

$$V = 2\pi r h\, \Delta r$$

and it can be remembered as

$$V = [\text{circumference}][\text{height}][\text{thickness}]$$

The Method of Cylindrical Shells

Now let S be the solid obtained by rotating about the y-axis the region bounded by $y = f(x)$ [where $f(x) \geqslant 0$], $y = 0$, $x = a$, and $x = b$, where $b > a \geqslant 0$. (See Figure 3.)

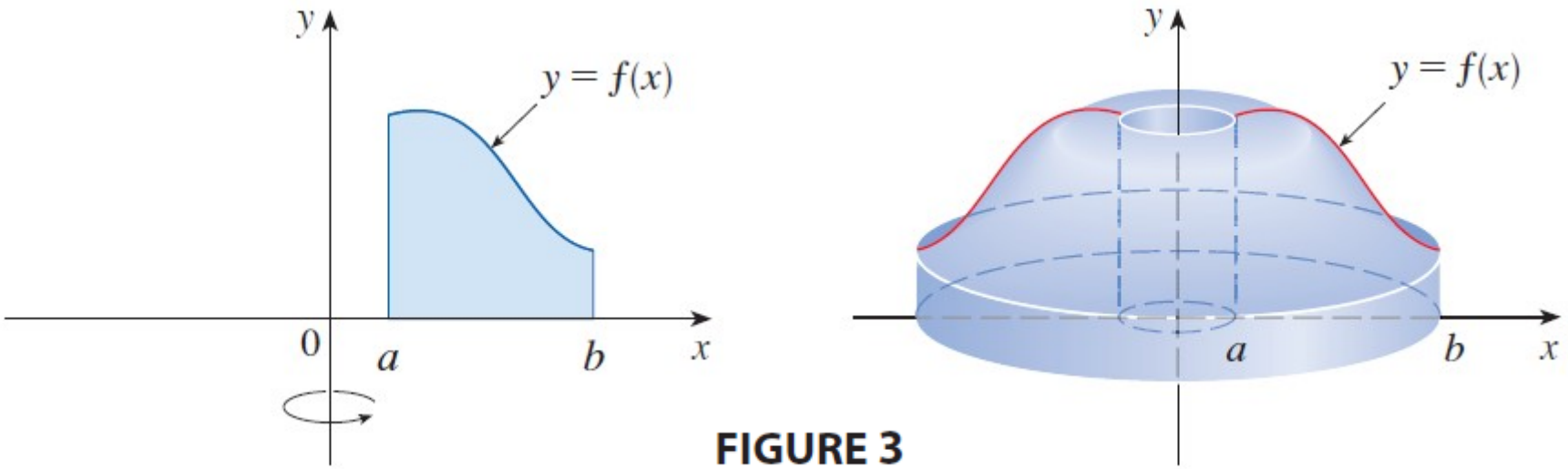


FIGURE 3

We divide the interval $[a, b]$ into n subintervals $[x_{i-1}, x_i]$ of equal width Δx and let $\bar{x}_i$ be the midpoint of the ith subinterval. If the rectangle with base $[x_{i-1}, x_i]$ and height $f(\bar{x}_i)$ is rotated about the y-axis, then the result is a cylindrical shell with average radius $\bar{x}_i$, height $f(\bar{x}_i)$, and thickness Δx. (See Figure 4.) So by Formula 1 its volume is

$$V_i = (2\pi\bar{x}_i)[f(\bar{x}_i)]\,\Delta x$$

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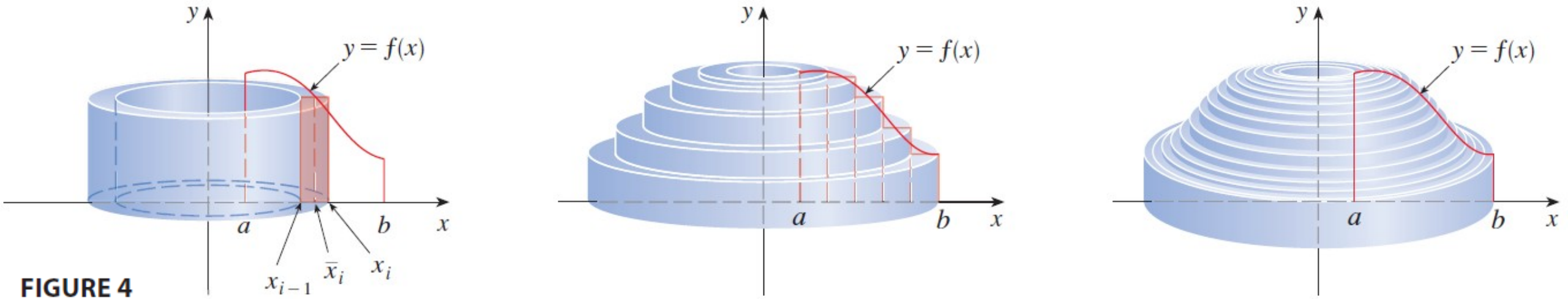


FIGURE 4

Therefore an approximation to the volume V of S is given by the sum of the volumes of these shells:

$$V \approx \sum_{i=1}^{n} V_i = \sum_{i=1}^{n} 2\pi\bar{x}_i f(\bar{x}_i)\,\Delta x$$

This approximation appears to become better as $n \to \infty$. But, from the definition of an integral, we know that

$$\lim_{n\to\infty} \sum_{i=1}^{n} 2\pi \bar{x}_i f(\bar{x}_i)\,\Delta x = \int_a^b 2\pi x\, f(x)\,dx$$

Thus the following formula appears plausible:

2 The volume of the solid in Figure 3, obtained by rotating about the y-axis the region under the curve $y = f(x)$ from a to b, is

$$V = \int_a^b 2\pi x f(x)\,dx \qquad \text{where } 0 \leqslant a < b$$

The best way to remember Formula 2 is to think of a typical shell, cut and flattened as in Figure 5, with radius x, circumference $2\pi x$, height $f(x)$, and thickness Δx or dx:

$$V = \int_a^b \underbrace{(2\pi x)}_{\text{circumference}} \underbrace{[f(x)]}_{\text{height}} \underbrace{dx}_{\text{thickness}}$$

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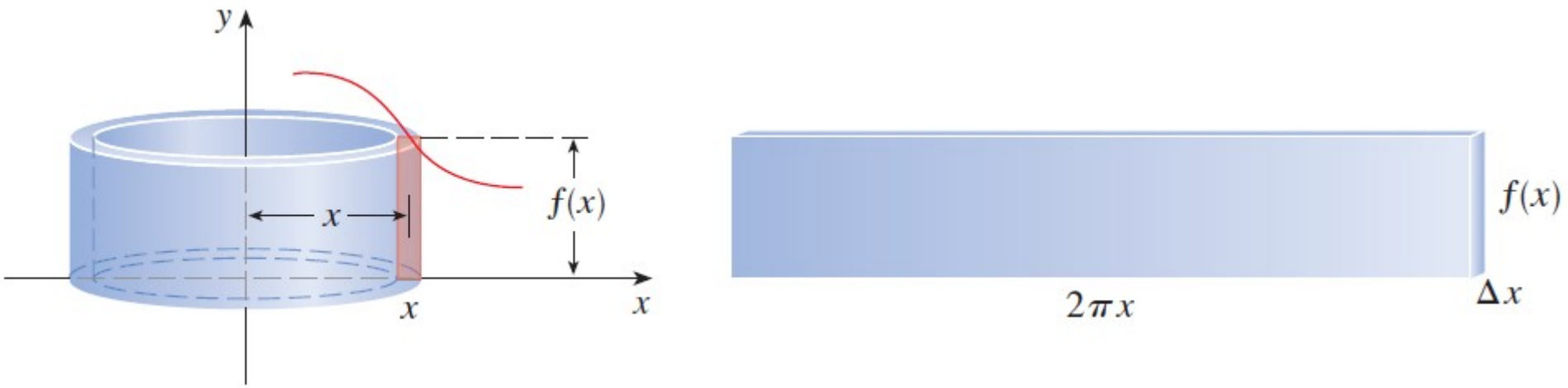
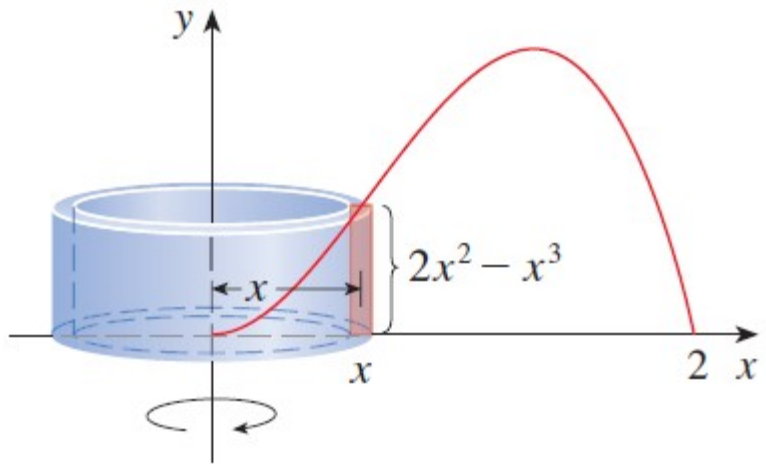


FIGURE 5

EXAMPLE

Find the volume of the solid generated by revolving the region bounded by the curves $y = 2x^2 - x^3$ and $y = 0$ about the y-axis.



SOLUTION

$$V = \frac{16}{5}\pi$$

EXAMPLE

Determine whether the **Disk/Washer method** or **Shell method** is better for finding the volume of the solid generated by revolving the region bounded by the curves $y = 2x^2 - x^3$ and $y = 0$ about the ***x*-axis**.

SOLUTION

EXAMPLE

Find the volume of the solid generated by revolving the region bounded by the curves $y = x^2$ and $y = x + 2$ about the line $x = 3$.

SOLUTION

$$V = \frac{45}{2}\pi$$

EXAMPLE

Find the volume of the solid formed by revolving the region bounded by the graph of

$$x = e^{-y^2}$$

and the y-axis $(0 \leq y \leq 1)$ about the x-axis.

SOLUTION

$$V = \pi(1 - e^{-1})$$

EXAMPLE

Find the volume of the solid generated by revolving the region bounded by the curves $x = 2y^2$ and $x = y^2 + 1$ about the line $y = -2$.

SOLUTION

$$V = \frac{16}{3}\pi$$

EXAMPLE

Find the volume of the solid generated by revolving the region bounded by the curves $y^2 - x^2 = 1$ and $y = 2$ about the y-axis.

SOLUTION

$$V = \frac{4}{3}\pi$$

Solve exercises of section 7.3.

Start with the suggested problems mentioned in the syllabus.